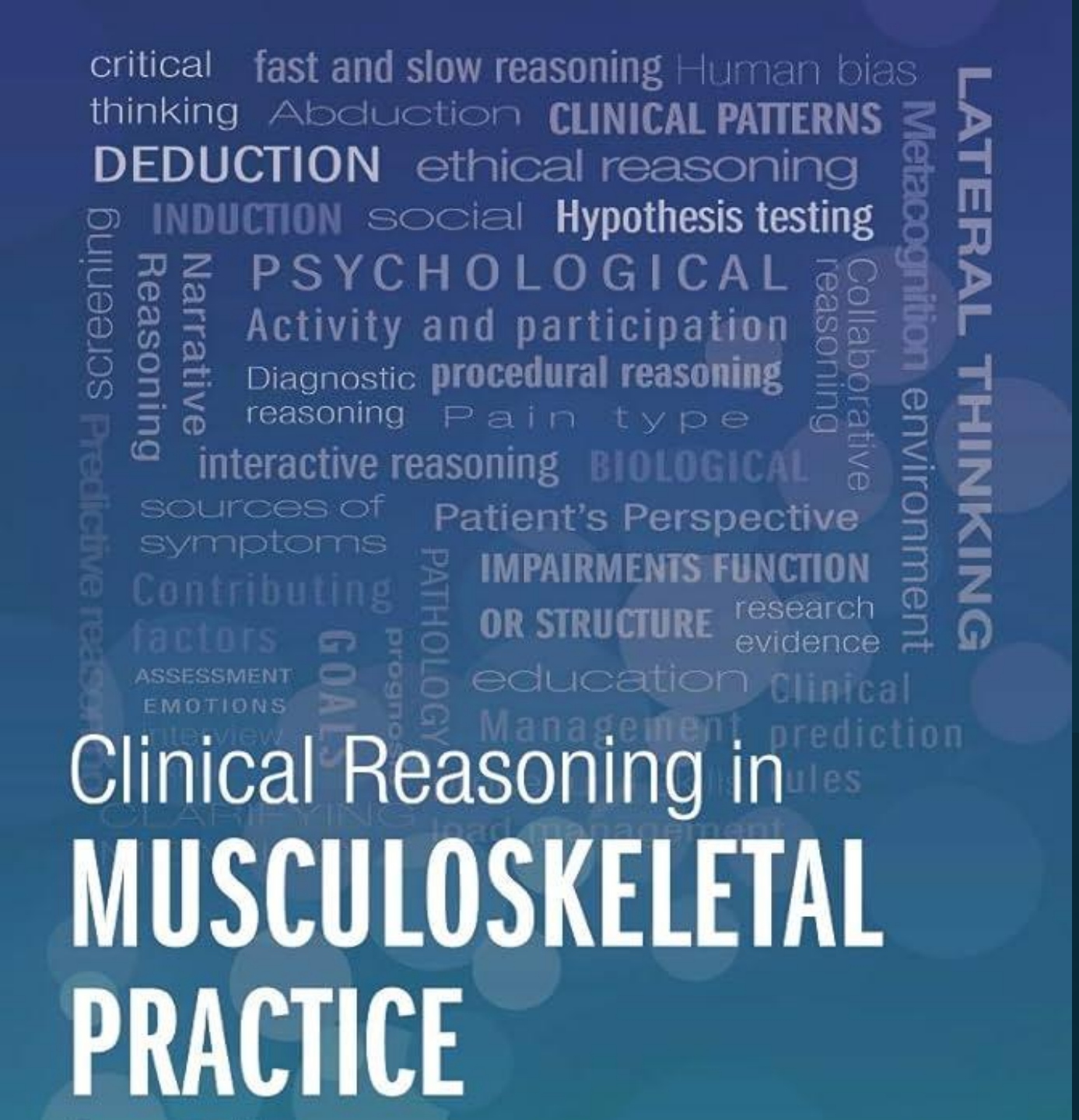


Clinical reasoning models in MSK

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Learning outcomes

Describe	Describe key clinical reasoning models in MSK physiotherapy and their strengths and limitations.
Apply	Apply selected reasoning models to MSK case studies to generate hypotheses and management options.
Integrate	Integrate UK guidance (NICE, HCPC) and evidence-based practice principles into decision-making.
Reflect on	Reflect on their developing reasoning skills and identify strategies to progress from beginner to advanced practice.

Introduction

- Definition of Clinical Reasoning: “The thinking and decision-making processes that physiotherapists use in clinical practice to gather and evaluate information, make diagnoses, and determine and implement interventions” (Higgs & Jones, 2008).
- Why it matters in MSK physiotherapy:
  - Helps make sense of complex patient presentations.
  - Guides safe and effective treatment.
  - Integrates evidence, patient preferences, and clinical expertise (EBP triad).

“A 40-year-old office worker with shoulder pain — what questions do you ask first? What models might you use?”

Models of Clinical Reasoning in MSK

1. Hypothetico-Deductive Model (pattern of hypothesis generation and testing).
2. Pattern Recognition Model (fast reasoning in familiar cases).
3. Forward Reasoning / Inductive model (data → conclusion).
4. Narrative Reasoning (understanding the patient's story/meaning of illness).
5. Pragmatic Reasoning (considering context: time, resources, patient goals).
6. Dual Process Theory (System 1 = fast, intuitive; System 2 = slow, analytical).
7. International Framework for Clinical Reasoning (IFCR) (structured and contemporary).

1- Hypothetico-Deductive Model

Definition: Clinical reasoning based on generating multiple hypotheses early, then testing them systematically through history-taking and examination (Elstein et al., 1978).

Example: Patient with knee pain — hypotheses might include meniscal injury, patellofemoral pain, OA. You ask targeted questions/tests to rule in/out.

Strengths: Structured, analytical, good for complex/novel cases.

Limitations: Time-consuming; risk of cognitive overload.

When to use each model

When to use the model	When not to use
When faced with unfamiliar or complex presentations.	Not efficient for straightforward or highly familiar cases.
Early in training, when you don’t yet recognise patterns.	Risk of “tunnel vision” if too many hypotheses distract from the main problem.
Useful in exams/OSCEs where systematic reasoning is rewarded.	At risk of confirmation bias (only looking for evidence that supports the hypothesis) and premature closure (settling too quickly on one possibility).

Case study (10 minutes)

- A 35-year-old recreational runner presents with 4 months of gradually worsening anterior knee pain. It started after she increased training mileage and now hurts with running, descending stairs and prolonged sitting.
- She reports occasional clicking but no true locking. There is no history of a specific trauma.
- Symptoms are worse at the end of the day, she can still run but only half her usual distance, and she is worried about whether continuing to train will “damage” her knee. No systemic symptoms.
- pain at the inferior patella and with a single-leg squat, reduced hip abductor strength on the symptomatic side, and aggravated pain on patellar compression.

How to use this technique

- Generate 4–6 competing hypotheses.
- For each hypothesis list the one or two most informative history questions and objective tests you would perform to rule in/out that hypothesis (be explicit about order).
- Decide which hypothesis to prioritise while you gather further data, and propose the next 2 immediate management decisions.
- Identify any red flags that would change your approach.

2- Pattern Recognition Model

Definition: Fast recognition of clinical patterns based on prior experience and knowledge (Norman et al., 2007).

MSK Example: A patient presents with pain radiating down the posterior leg, numbness in the lateral foot, and weakness in plantarflexion → clinician immediately recognizes this as an S1 radiculopathy without consciously working through each step.

Strengths: Efficient, fast useful in experienced clinicians.

Limitations: high risk of bias, premature closure, less reliable in atypical cases.

Case Study 2 (5 minutes)

- A 45-year-old administrative assistant develops a 6-week history of lateral elbow pain. It began gradually after a period of heavy gardening. She reports pain when turning a screwdriver, pouring from a kettle and lifting a shopping bag. Pain is mainly local and focal over the lateral epicondyle, worse with gripping, and she notices weakness when shaking hands. No neck pain, no numbness or systemic signs.
- She reported pain on resisted wrist extension and with resisted middle-finger extension, provocative gripping reproduces symptoms.
- **State the rapid pattern diagnosis you would make and why (what cluster of features supports it).**
- **Outline the immediate management you would offer in a first 2–4 week plan (practical, time-limited).**

When to use	When not to use
In common conditions with classic presentations (e.g., tennis elbow, frozen shoulder).	Avoid if the case is atypical, complex, or has red flags.
Later in training when you've built up a bank of cases.	Risk of bias (anchoring, premature closure).

- begin to recognise common MSK patterns but always cross-check with objective testing.
- With experience: integrate guidelines (e.g., NICE frozen shoulder recommendations) to confirm pattern recognition is evidence-based.

3- Forward Reasoning (Inductive Reasoning)

- **Definition**: Reasoning from clinical data (symptoms, signs) directly to diagnosis without generating multiple hypotheses (Eva, 2005).
- **MSK Example**: Acute inversion ankle injury → swelling, bruising, lateral tenderness → likely lateral ligament sprain.
- **Strengths**: Efficient in familiar cases.
- **Limitations**: May miss red flags or alternative causes.

Case study (5 minutes)

- A 22-year-old male footballer twisted his ankle during a match 2 hours ago. Immediate swelling and bruising developed around the lateral malleolus; he cannot weight-bear without severe pain. He reports hearing a “pop” at the time. On inspection there is marked swelling and ecchymosis; he cannot take four steps unassisted. No other injuries.

  - **Starting from the data points provided, work forward to the most likely diagnoses in order of probability.**
  - **State which immediate bedside decisions you would make (immobilise? x-ray? analgesia?).**
  - **List the quick objective tests you would perform on first contact**

When to use this model

When to use the model	When not to use
Acute injuries with clear mechanism and signs (e.g., ankle sprains, fractures).	Avoid overconfidence — not suitable for subtle or chronic conditions where multiple possibilities exist.
When validated clinical rules exist (e.g., **Ottawa Ankle Rules**, recommended by NICE CG176 Fractures).	Risks missing underlying pathology if reasoning is too linear.
InOSCE students are expected to generate differential diagnoses and then use clinical tests to	

How to use these models in practice

Eg: In case of a patient with ACL:

- You might **recognize** a classic ACL tear from the story,
- then use **deductive reasoning** to test Lachman and pivot shift,
- and if the presentation is unclear, switch to **inductive reasoning** to carefully build the picture.



Narrative Reasoning

Definition: Understanding the patient's experience, beliefs, and story to inform care (Mattingly, 1991).

MSK Example: Chronic low back pain — patient links pain to work stress, fears movement → reasoning helps tailor education and graded activity.

Strengths: Patient-centred, supports shared decision-making.

Limitations: Less structured; requires good communication skills.

Case Study

- A 50-year-old warehouse operative presents with a 2-year history of low back pain. He was off work for 3 months after the initial episode and returned on light duties.
- He says, “I’m scared to lift now — I don’t trust my back. My wife nags me about it and I can’t sleep; I worry I’ll slip up and lose my job.”
- He expresses frustration that clinicians “just give exercises” and that nothing addresses why he keeps getting worse.
- He is low on sleep and reports feeling down.
- **Use this case to practise empathic listening, goal-setting aligned with work participation, and linking physical rehab to function and meaning.**

When to use this model

- begin practicing open questions, active listening, and understanding “yellow flags.”
- With experience: integrate narrative reasoning with **shared decision-making** (NICE encourages collaborative care planning).

When to use the model	When not to use
Chronic conditions where psychosocial context affects recovery (e.g., persistent low back pain, whiplash, fibromyalgia). When patient beliefs/fears are major barriers.	Not sufficient alone must be combined with diagnostic reasoning. May become unfocused without structure.

Pragmatic Reasoning

Definition: Considering practical aspects like time, resources, environment, and patient preferences in decision-making (Higgs & Jones, 2008).

Example: Deciding to prescribe home-based exercises instead of supervised gym program due to patient's limited availability.

Strengths: Realistic, adaptable.

Limitations: May conflict with ideal best-practice guidelines.

Case Study (7 minutes)

- A 28-year-old new mother (6 weeks postpartum) presents with radial wrist/ thumb pain consistent with de Quervain's tenosynovitis.
- She is exclusively breastfeeding, has a 5-month-old toddler to care for, no partner available during the day, and limited funds for frequent outpatient sessions.
- She wants fast relief because she needs to carry the baby and shop independently.
- **Prioritise three achievable goals for the first 4 weeks given her constraints**

When to use this model

NICE encourages self-management and pragmatic pathways (e.g., home exercise, group classes for OA).
HCPC requires clinicians to adapt practice to context while maintaining professional standards.

When to use the model	When not to use
When practical barriers affect treatment choice (time, cost, access). NHS or community settings where resources are limited.	Avoid if it leads to substandard care or disregards best evidence. Must not compromise safety.

Dual Process Theory (System 1 & 2 Thinking)

Definition: Two modes of reasoning — fast, intuitive (System 1) and slow, analytical (System 2) (Kahneman, 2011).

Experienced physio quickly recognising frozen shoulder (System 1), but switching to System 2 to carefully rule out red flags.

Strengths: Explains how both experience and analysis contribute to reasoning.

Limitations: Risk of cognitive bias when relying too much on System 1.

Case study

- A 60-year-old woman presents with a 3-month history of progressive shoulder stiffness and pain.
- She reports difficulty reaching overhead, reduced sleep due to achiness, and increasing difficulty dressing.
- On examination passive and active external rotation are substantially reduced, abduction is limited; rotator cuff strength is hard to assess because of pain but grossly present. She has a history of type 2 diabetes.
- **What is your system 1 says and what would your system 2 considers?**

When to use this model

consciously practise both — make an intuitive guess, then deliberately test alternatives.
With experience: intuition becomes more reliable, but clinicians must still pause and check.

When to use the model	When not to use
Always in practice — clinicians naturally switch between fast and slow reasoning. Useful for safety: start with intuition (pattern recognition) but check with analysis (rule out red flags).	Over-reliance on System 1 → cognitive biases, misdiagnosis. Over-reliance on System 2 → inefficiency in busy clinics.

International Framework for Clinical Reasoning (IFCR)

Definition: A comprehensive framework describing reasoning processes across contexts, including diagnostic, narrative, interactive, and collaborative reasoning (Ajjawi et al., 2020).

MSK Example: Applying multiple reasoning approaches in a complex case like persistent knee pain post-ACL reconstruction.

Strengths: Integrative, modern, evidence-informed.

Limitations: Broad — may be harder for early students to grasp fully.

When to use this model

When to use the model	When not to use
Complex cases requiring integration of multiple reasoning types (diagnostic, narrative, collaborative, ethical).	Can overwhelm students if applied too early — too broad and abstract.
MDT and return-to-sport scenarios.	

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